



Digital Transmission

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Digital-to-Digital Transmission

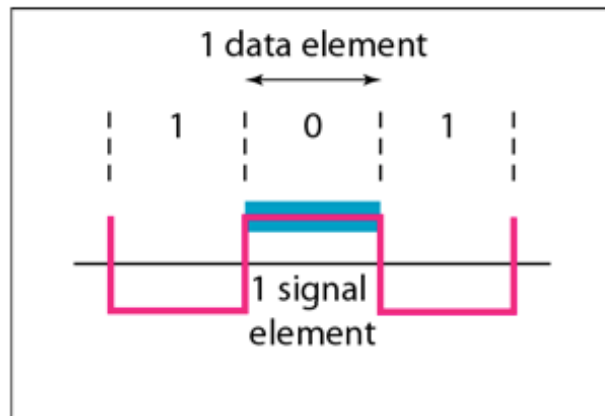
- Three techniques
 - Line coding
 - Block coding
 - Scrambling



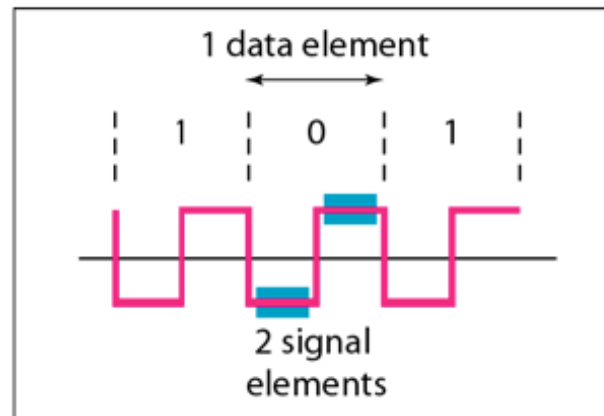
Data element and Signal element

- Converts a sequence of bits to a digital signal
- A **data element** is the smallest entity that represent a piece of information; this is the bit.
- A **signal element** is the shortest unit (timewise) of a digital signal.
- Data elements are being carried
- Signal element are the carriers
- A ratio r defines number of data elements carried by each signal element

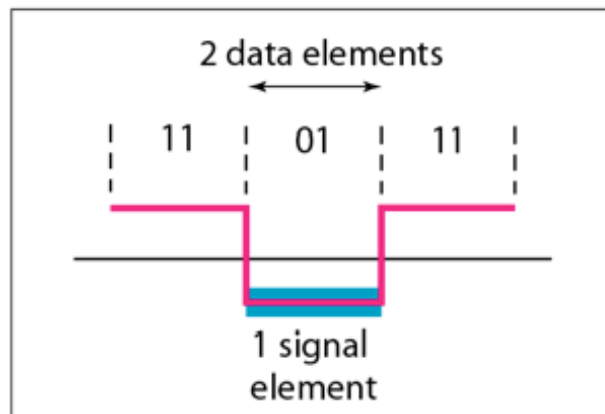
Data element and Signal element



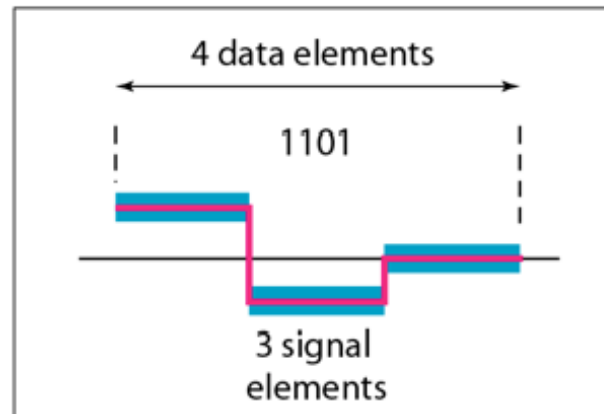
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)



Bit rate vs Baud rate

- **Data rate** (bit rate) defines the number of data elements (bit) sent in 1s. The unit is bits per second (**bps**)
- The **signal rate** (baud rate) is the number of signal elements sent in 1s. The unit is **baud**.
- Goal is to increase the data rate while decreasing the signal rate
- Relationship between data rate and signal rate as
 - $S = c \times N \times 1/r$ baud
- where N is the data rate (bps), c is the case factor, which varies for each case, S is the number of signal elements, and r is the previously defined factor



Example

- A signal is carrying data in which one data element is encoded as one signal element ($r = 1$). If the bit rate is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?
- We assume that the average value of c is $1/2$. The baud rate is then

$$S = c \times N \times 1/r$$

$$= 1/2 \times 100,000 \times 1 = 50,000 = 50 \text{ kbaud}$$



Minimum Bandwidth

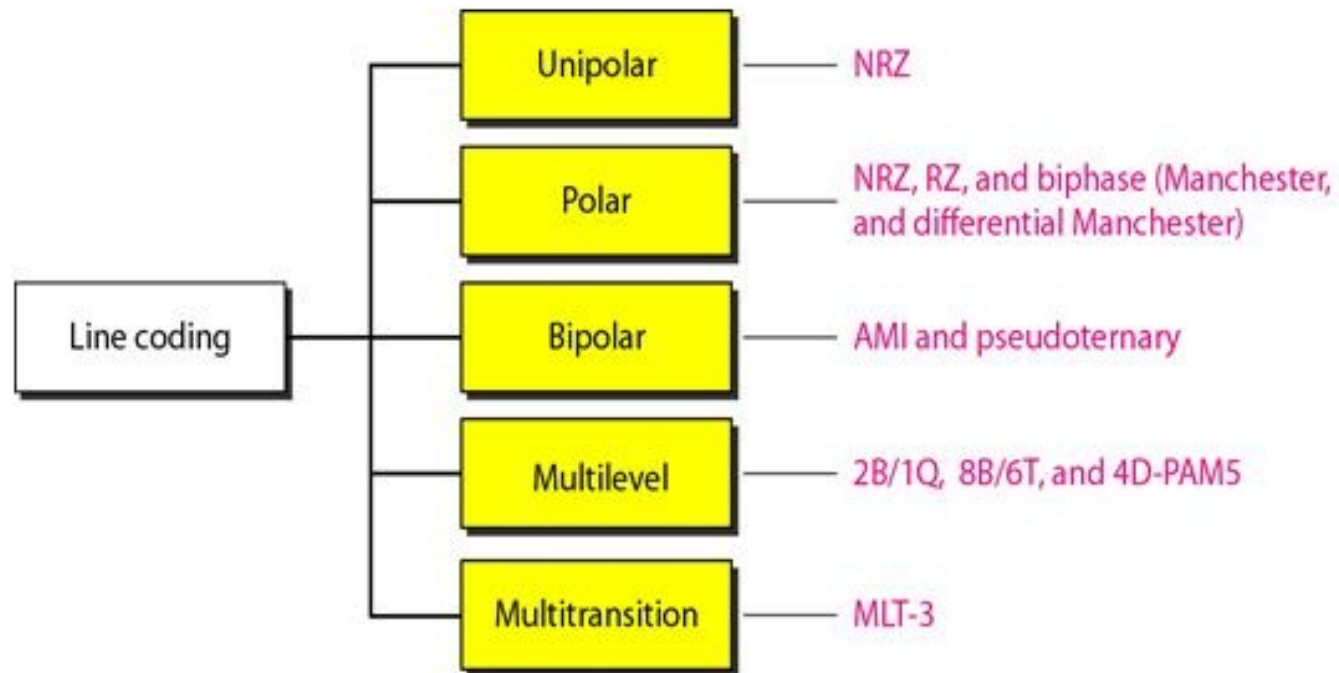
- Baud rate determines the required bandwidth for a digital signal
- Bandwidth is proportional to the signal rate (baud rate)
- The minimum bandwidth can be given as
 - $B_{\min} = c \times N \times 1/r$ baud
- Maximum data rate
 - $N_{\max} = 1/c \times B_{\min} \times r$



Need of encoding scheme

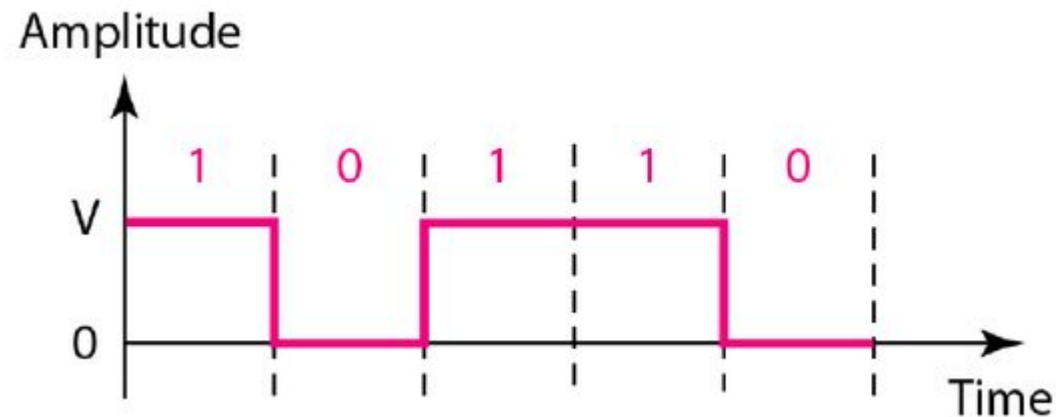
- Baseline Wandering
 - A long string of 0s or 1s can cause a drift in the baseline and make it difficult for the receiver to decode correctly
- DC (direct-current) components
 - Constant voltage level creates very low frequencies around zero
- Self-synchronizing digital signal
 - The clock speed of receiver and sender may not match
- Built-in Error Detection
 - Handling small errors via encoding
- Immunity to Noise and Interference
- Complexity

Line Coding Schemes



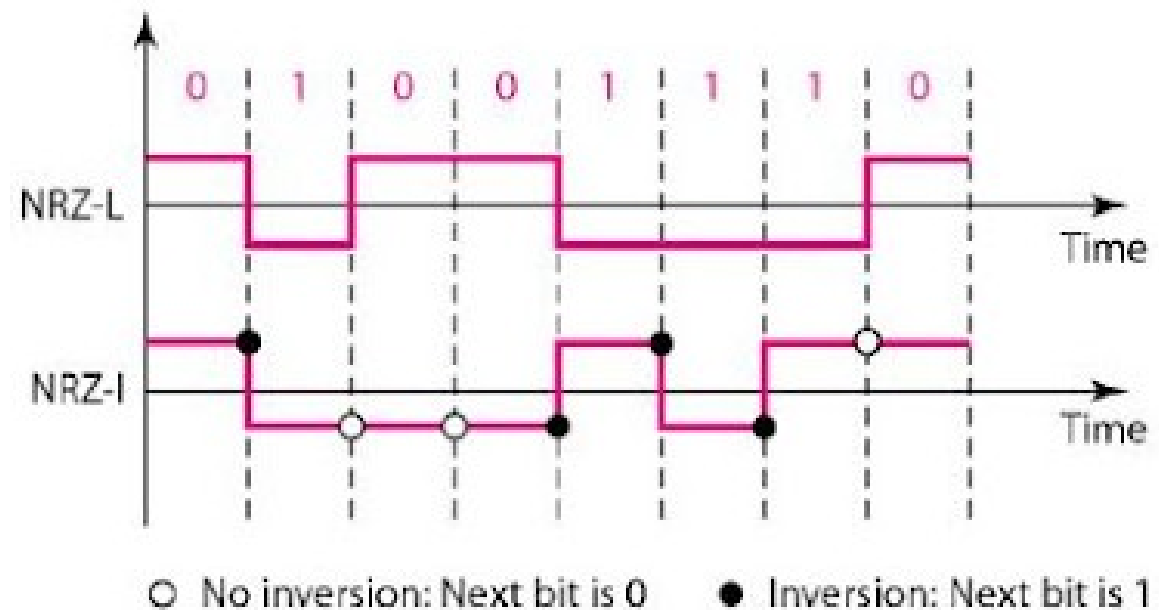
Unipolar Scheme

- In a unipolar scheme, all the signal levels are on either above or below of the time axis.
- **NRZ (Non-Return-to-Zero)**
 - The positive voltage defines bit 1 and the zero voltage defines bit 0. It is called NRZ because the signal does not return to zero at the middle of the bit.



Polar scheme

- The voltages are on the both sides of the time axis
 - Example, voltage level for 0 can be positive and the voltage level for 1 can be negative
- Non-Return-to-Zero (NRZ)
 - In polar NRZ encoding, Two levels of voltage amplitude is used with two variations (NRZ-L and NRZ-I)



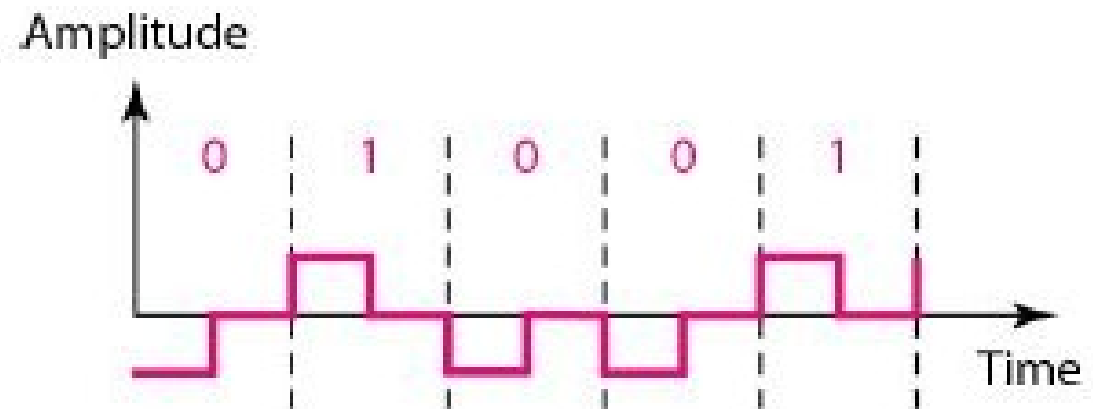


NRZ-L vs NRZ-I

- Baseline wandering is twice as severe in NRZ-L
- The synchronization problem also exists in both schemes
- Sudden change of polarity problem in NRZ-L
- Both schemes have an average signal rate of $N/2 B_d$
- DC component problem persists

Return to Zero (RZ)

- Uses three values: positive, negative, and zero
- In RZ, the signal changes not between bits but **during the bit**
- The signal goes to 0 in the middle of each bit and it remains there until the beginning of the next bit
- It requires two signal changes to encode a bit
 - Complex and occupies greater bandwidth



Polar Biphase : Manchester and Differential Manchester schemes



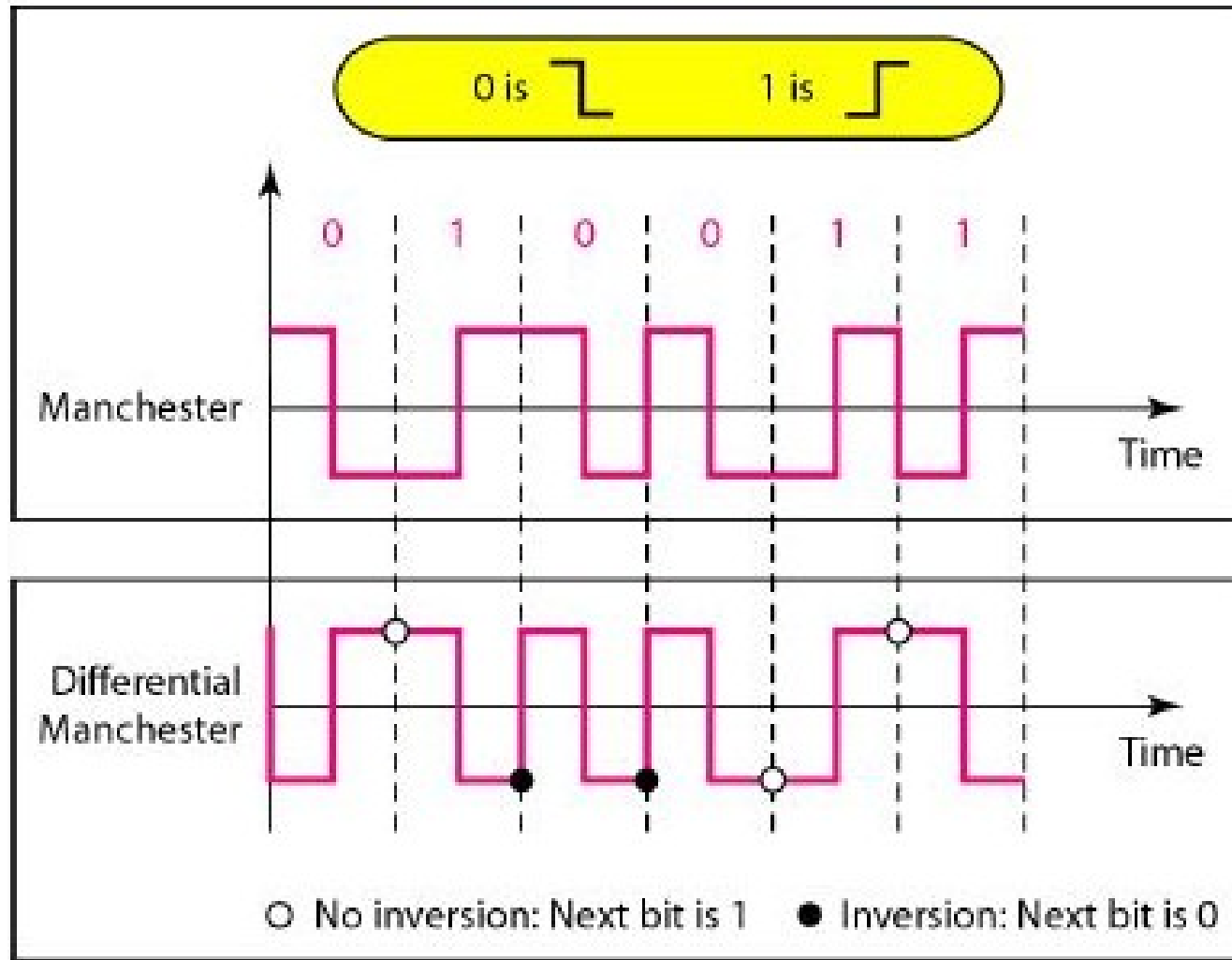
- **Manchester**

- combines the idea of RZ and NRZ-L
- The duration of the bit is divided into two halves
- The voltage remains at one level during the first half and moves to the other level in the second half
- The transition at the middle of the bit provides synchronization

- **Differential Manchester**

- combines the ideas of RZ and NRZ-I
- There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit
- If the next bit is 0, there is a transition; if the next bit is 1, there is none

Manchester and Differential Manchester schemes



Manchester and Differential Manchester schemes(2)



- The Manchester scheme has no baseline wandering, no DC component because each bit has a positive and negative voltage contribution
- The only drawback is the signal rate, which is double compared to NRZ
 - Because, there is always one transition at the middle of the bit and maybe one transition at the end of each bit

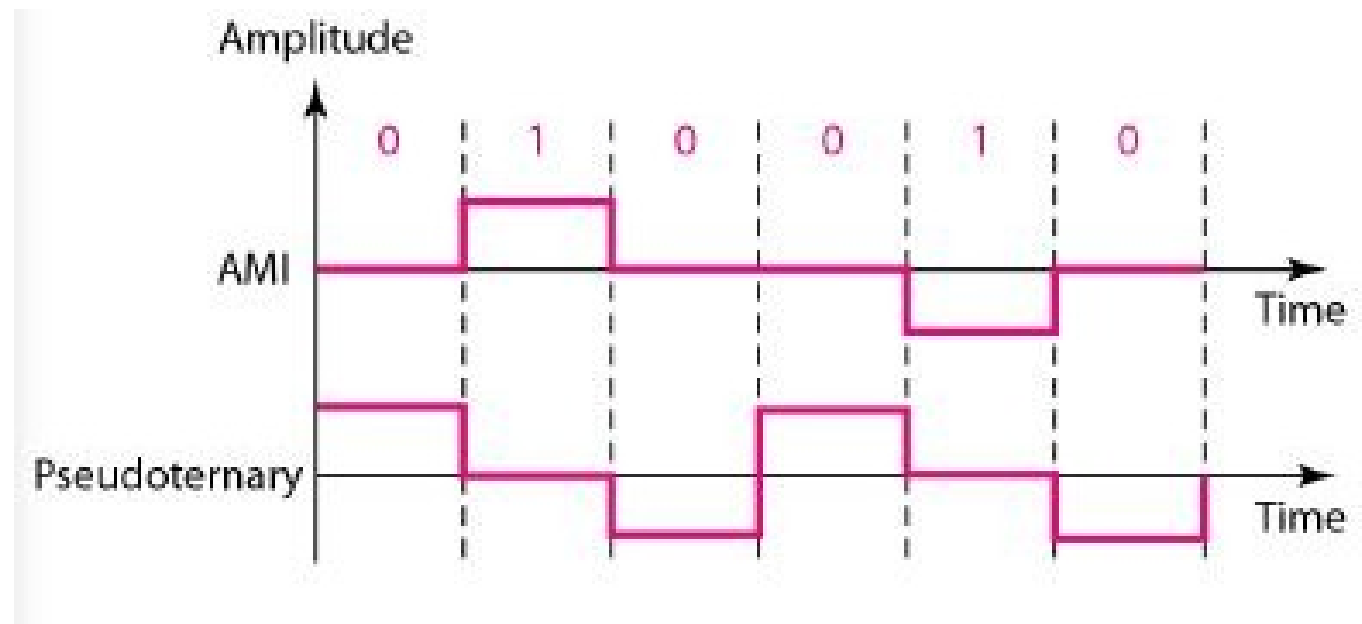


Bipolar Schemes

- In bipolar encoding (sometimes called multilevel binary), there are three voltage levels: positive, negative, and zero
- The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative
- A common bipolar encoding scheme is called bipolar **alternate mark inversion (AMI)**
 - The word mark comes from telegraphy and means 1
 - AMI means alternate 1 inversion
 - A neutral zero voltage represents binary 0
 - Binary is are represented by alternating positive and negative voltages

Pseudoternary

- A variation of AMI encoding is called **pseudoternary** in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages

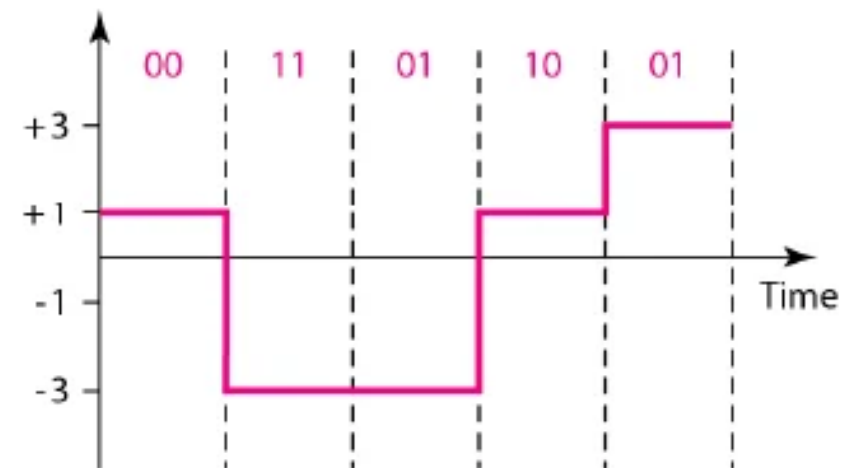


Multilevel Schemes : 2B1Q

- 2B1Q (two binary, one quaternary)
- Uses data patterns of size 2 and encodes the 2-bit patterns as one signal element belonging to a four-level signal

	Previous level: positive	Previous level: negative
Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

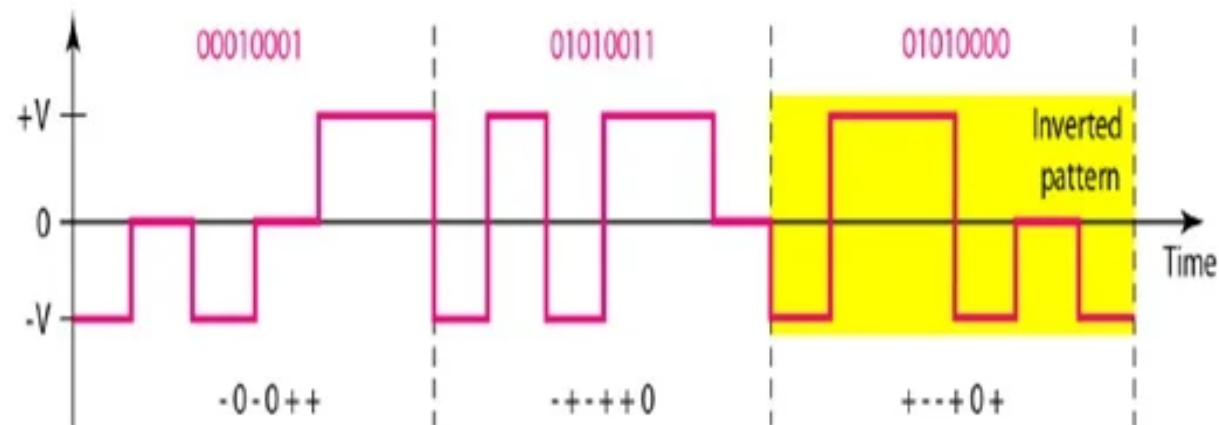
Transition table



Assuming positive original level

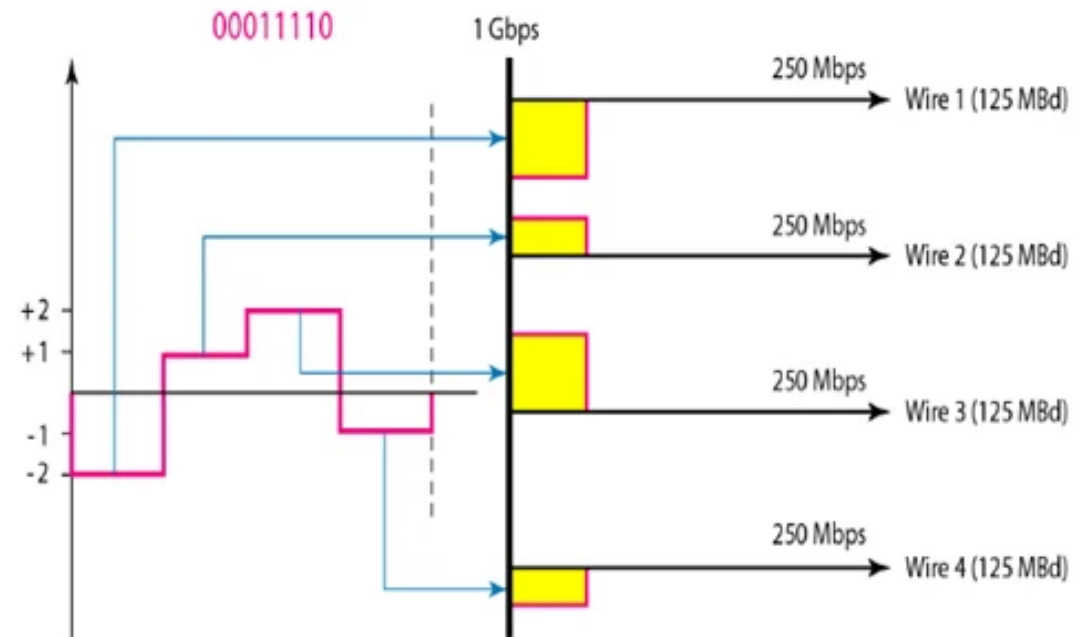
Multilevel Schemes : 8B6T

- Encode a pattern of 8 bits as a pattern of 6 signal elements
- 3 signal levels are used : 0, -V, and +V
- $2^8 = 256$ different data patterns and $3^6 = 478$ different signal patterns
- A mapping table is defined where $478 - 256 = 222$ redundant signal elements present that provide synchronization and error detection
- The sum of the signal elements for a single pattern is called weight of the signal element



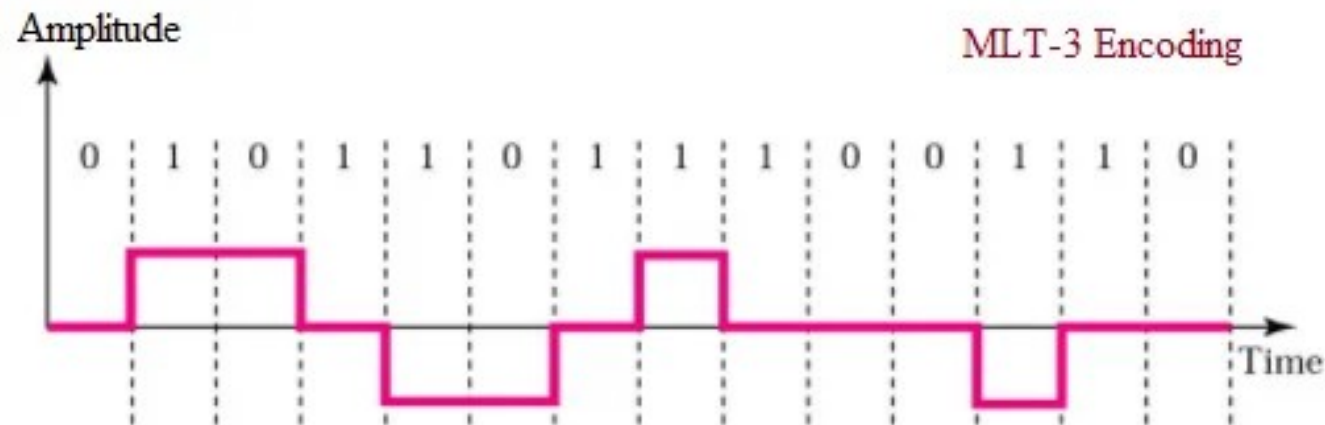
Multilevel Schemes : 4D-PAMS

- Four-dimensional five-level pulse amplitude modulation (4D-PAMS)
- 4D means that data is sent over four wires at the same time
- It uses five voltage levels, such as -2, -1, 0, 1, and 2.
- However, one level, level 0, is used only for forward error detection



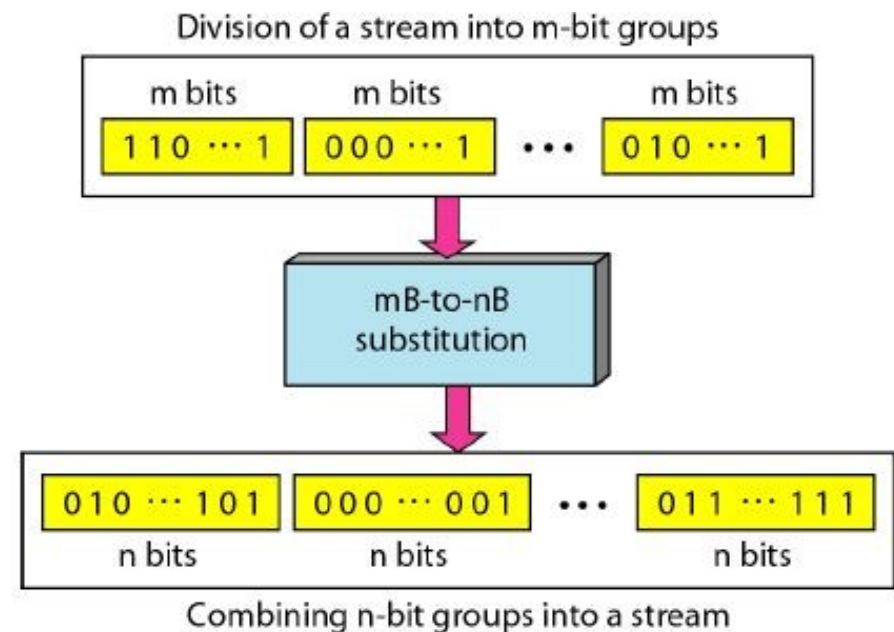
Multiline Transmission: MLT-3

- Uses three levels (+ V, 0, and - V) and three transition rules to move between the levels
- If the next bit is 0, there is no transition
- If the next bit is 1 and the current level is not 0, the next level is 0
- If the next bit is 1 and the current level is 0, the next level is the opposite of the last nonzero level



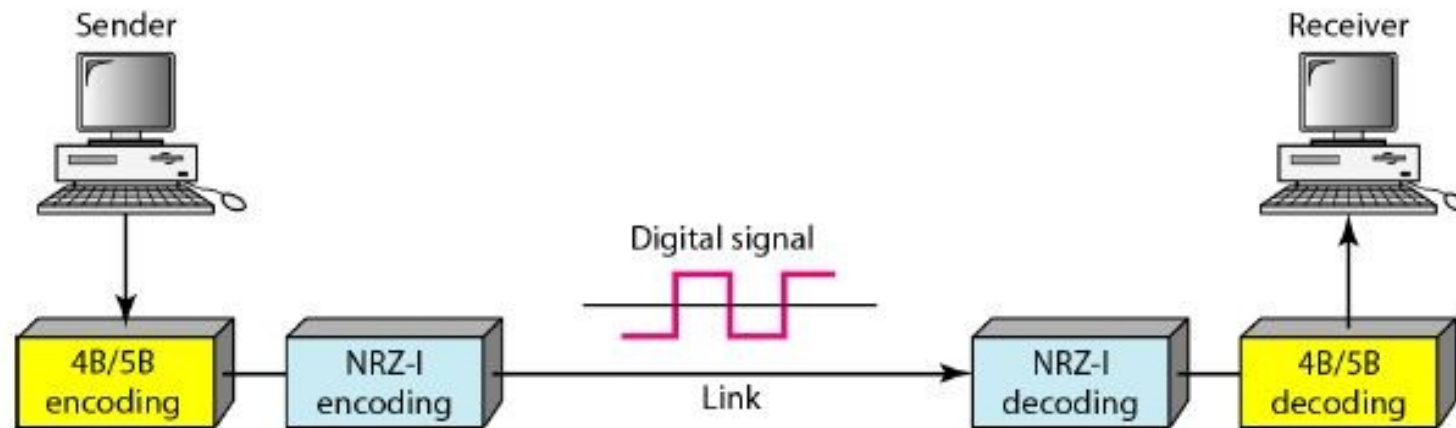
Block Encoding

- Block coding can give us redundancy and improve the performance of line coding
- Block coding changes a block of m bits into a block of n bits, where n is larger than m
- Block coding is referred to as an mB/nB
- For example, in 4B/5B encoding we substitute a 4-bit code for a 5-bit group



4B/5B

- Designed to be used in combination with NRZ-I



4B/5B

- Designed to be used in combination with NRZ-I

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

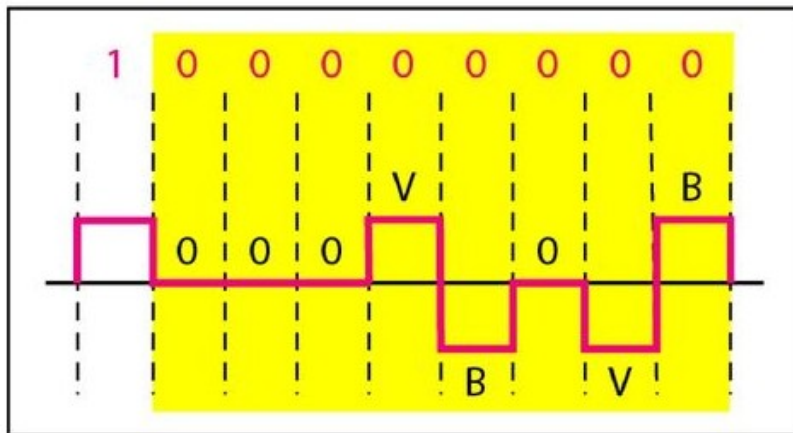


Scrambling

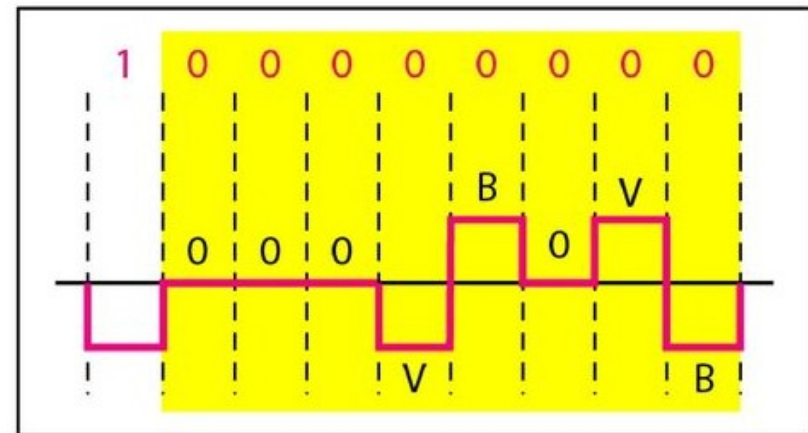
- Biphase schemes, block coding and NRZ line coding **are not suitable** for **long-distance communication**
- Bipolar AMI encoding is good but a **long sequence of 0s upsets the synchronization**
- We are looking for a solution that substitutes long zero-level pulses with a combination of other levels to provide synchronization
- One solution is called **scrambling**
- It is done at the same time as encoding
- The system needs to insert the required **pulses** based on the defined **scrambling rules**
- Two common scrambling techniques are B8ZS and HDB3

B8ZS

- Bipolar with 8-zero substitution (B8ZS)
- Eight consecutive zero-level voltages are replaced by the sequence OOOVBOVB
 - Where, the V denotes violation; this is a nonzero voltage that breaks an AMI rule of encoding
 - The B in the sequence denotes bipolar which means a nonzero level voltage in accordance with the AMI rule
- The letter V (violation) or B (bipolar) here is relative. The V means the same polarity as the polarity of the previous nonzero pulse; B means the polarity opposite to the polarity of the previous nonzero pulse



a. Previous level is positive.



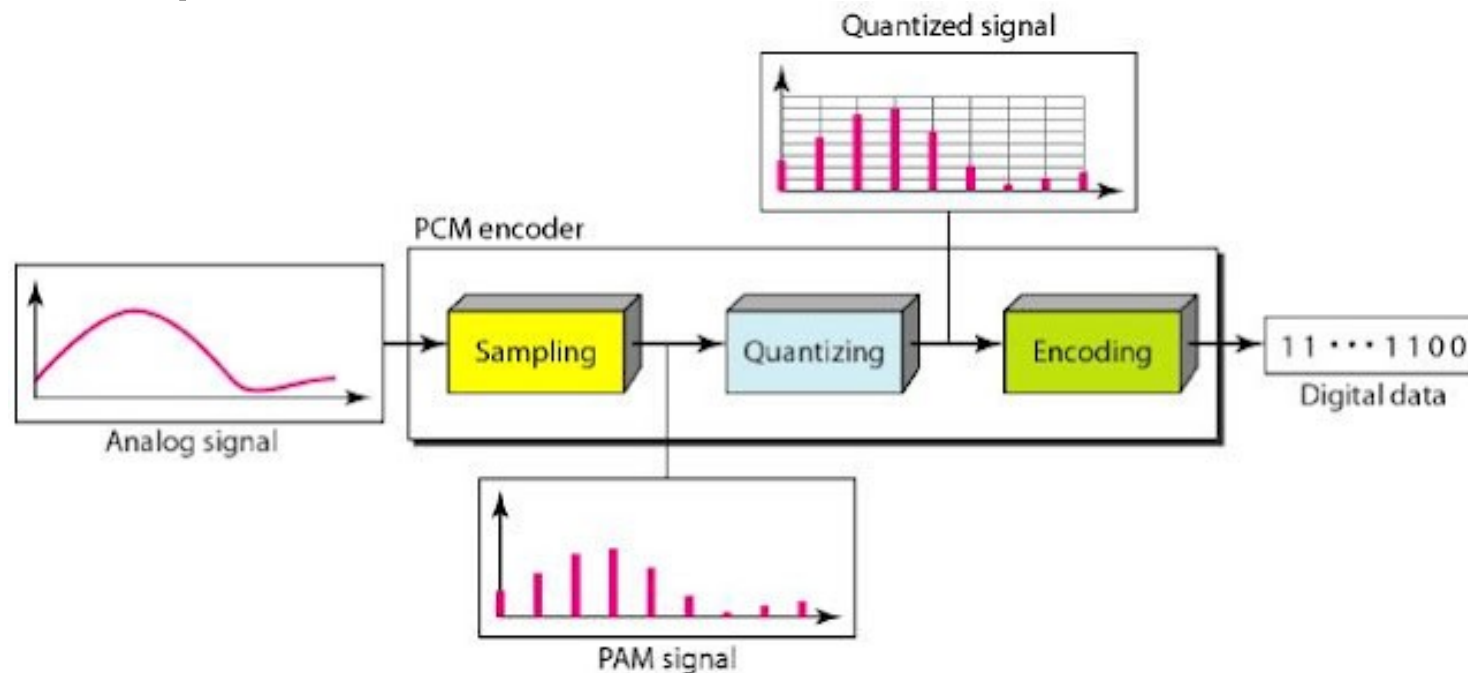
b. Previous level is negative.



ANALOG-TO-DIGITAL CONVERSION

Pulse Code Modulation (PCM)

- Consists of 3 major steps
 - The analog signal is **sampled**.
 - The sampled signal is **quantized**.
 - The quantized values are **encoded** as streams of bits.

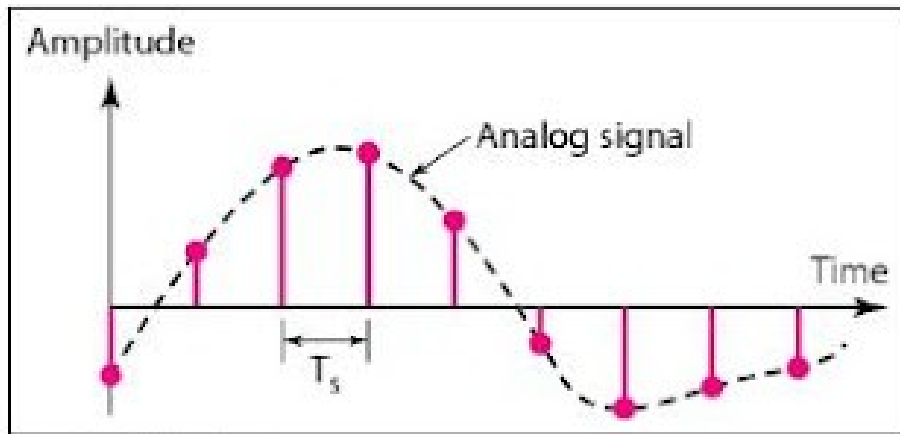




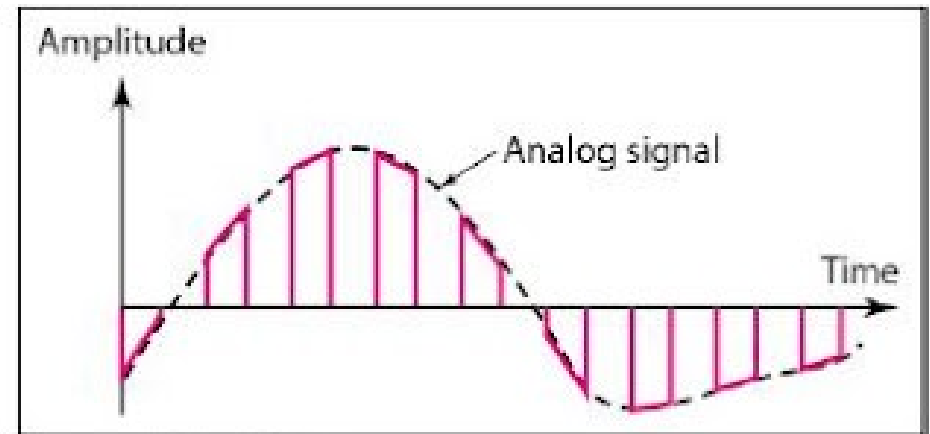
PCM First Step : Sampling

- The analog signal is measured for the instantaneous values of the continuous signal at regular intervals, of T_s sec, where T_s is the **sample interval** or period
- The inverse of the sampling interval is called the **sampling rate** or **sampling frequency** $f_s = 1/T_s$
- There are 3 sampling methods : ideal, natural, and flat-top
- **Ideal sampling**
 - Pulses from the analog signal are sampled
 - Cannot be easily implemented
- **Natural sampling**
 - A high-speed switch is turned on for only the small period of time when the sampling occurs
 - The result is a sequence of samples that retains the shape of the analog signal
- **Sample and hold**
 - Creates flat-top samples by using a circuit.

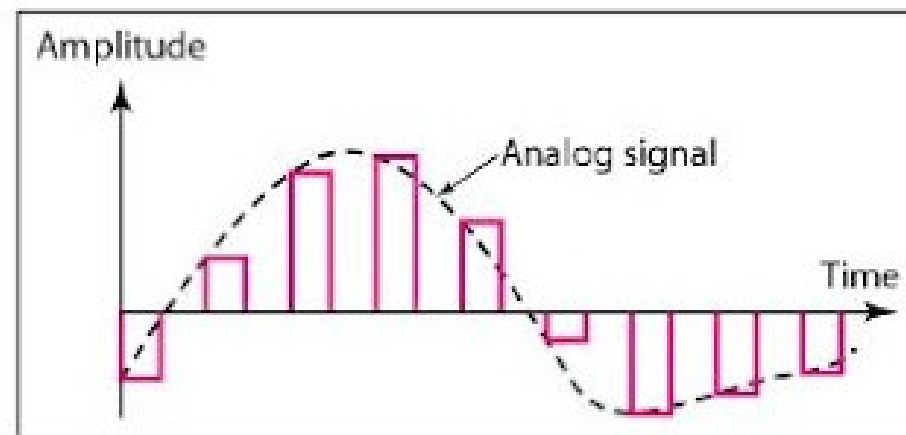
PCM First Step : Sampling



a. Ideal sampling



b. Natural sampling



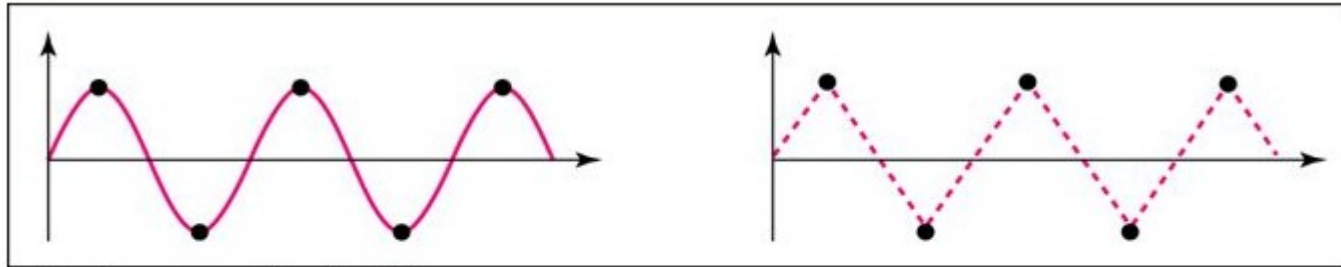
c. Flat-top sampling



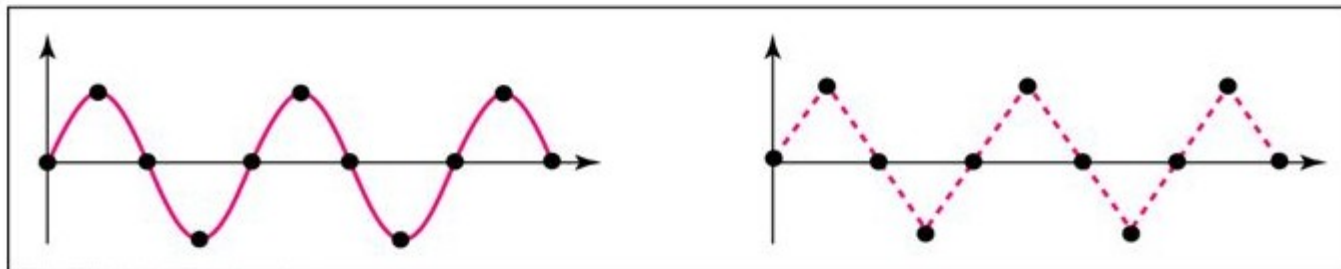
Sampling Rate (frequency of sampling)

- What are the restrictions on T_s ?
- Nyquist answered this beautifully
- According to the Nyquist theorem, to reproduce the original analog signal, the sampling rate be **at least twice** the **highest frequency** in the original signal
 - Sampling rate = $2 \times f_{\max}$

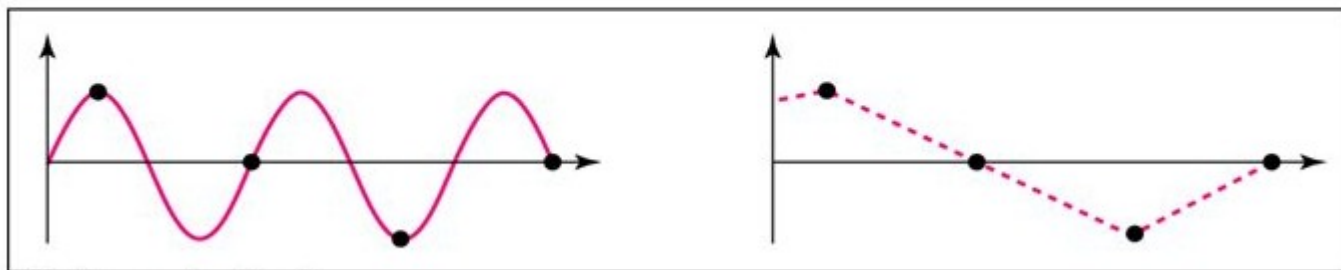
Sampling rate comparison



a. Nyquist rate sampling: $f_s = 2f$



b. Oversampling: $f_s = 4f$



c. Undersampling: $f_s = f$

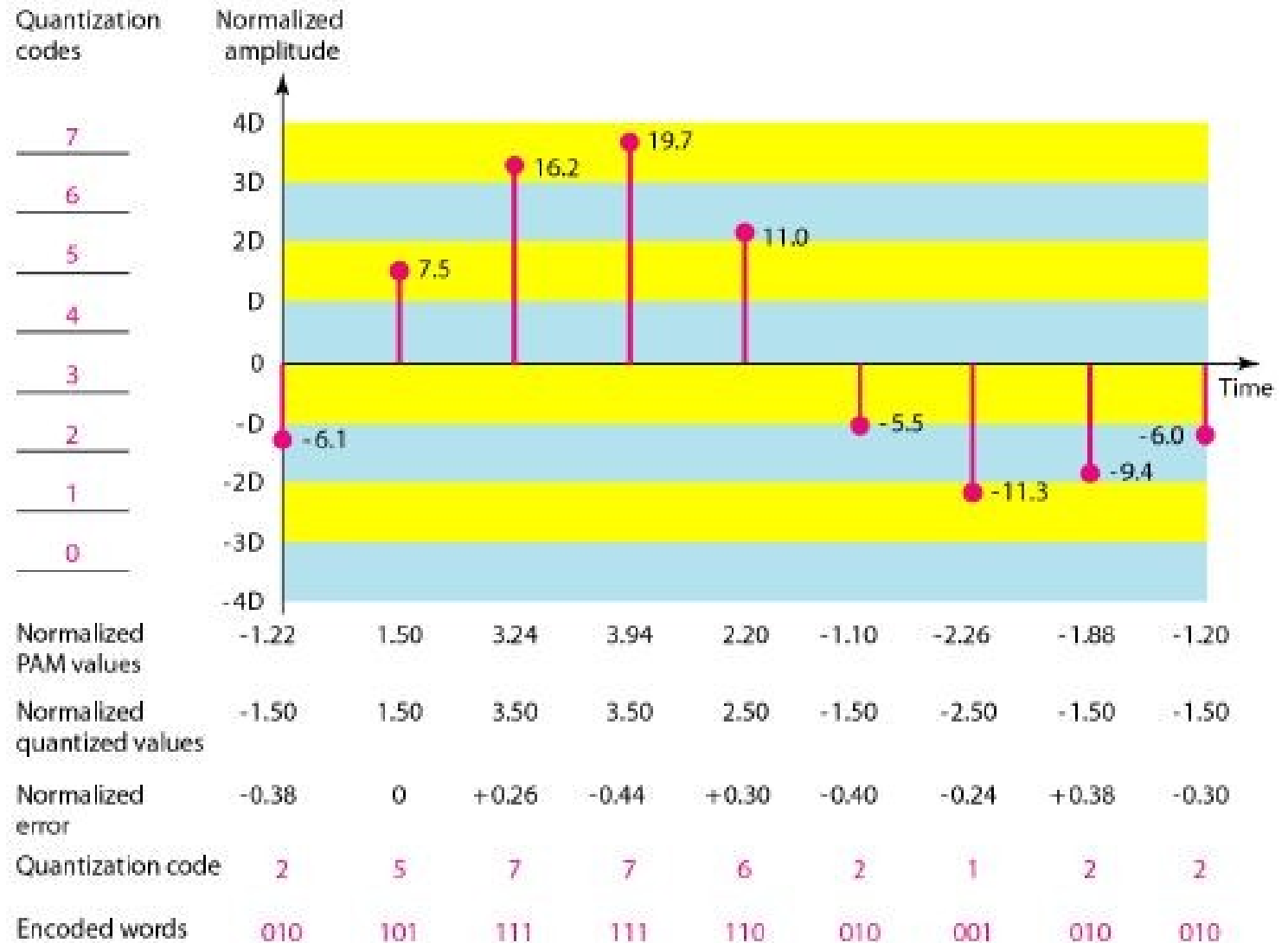


PCM Second step : Quantization

- Output of sampling is a series of pulses with amplitude
- The set of amplitudes can be infinite with nonintegral values between **maximum** and **minimum** limits
- To make these values useful in encoding the **quantization** is necessary
- THE PROCESS
 - Assume that the original analog signal has instantaneous amplitudes between $V_{\min}$ and $V_{\max}$
 - Divide the range into **L zones**, each of height Δ (delta).
 - $\Delta = (V_{\max} - V_{\min}) / L$
 - Assign quantized values of 0 to $L - 1$ to the midpoint of each zone.
 - **Approximate** the value of the sample amplitude to the quantized values

Second step : Quantization

$V_{\min} = -20 \text{ V}$
 $V_{\max} = +20 \text{ V}$
 $L = 8$
 $\Delta = 5 \text{ V}$
 Norm Amp = actual amplitude/ Δ





PCM Third step : Encoding

- Finalizing number of bits per sample
- Each sample can be changed to an $(\log_2 L)$ -bit code word
- In previous example, encoded words are shown in the last row
- The bit rate can be found from the formula
 - Bit rate = sampling rate x number of bits per sample



Other form of modulation

- Delta modulation (left for self reading)

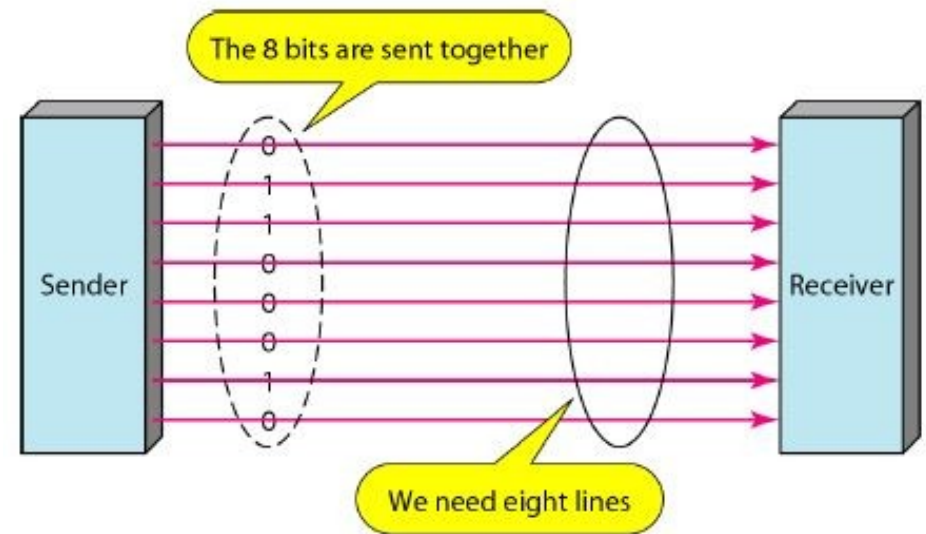


Transmission Mode

- Parallel Transmission
- Serial Transmission
 - Asynchronous Transmission
 - Synchronous Transmission
 - Isochronous Transmission

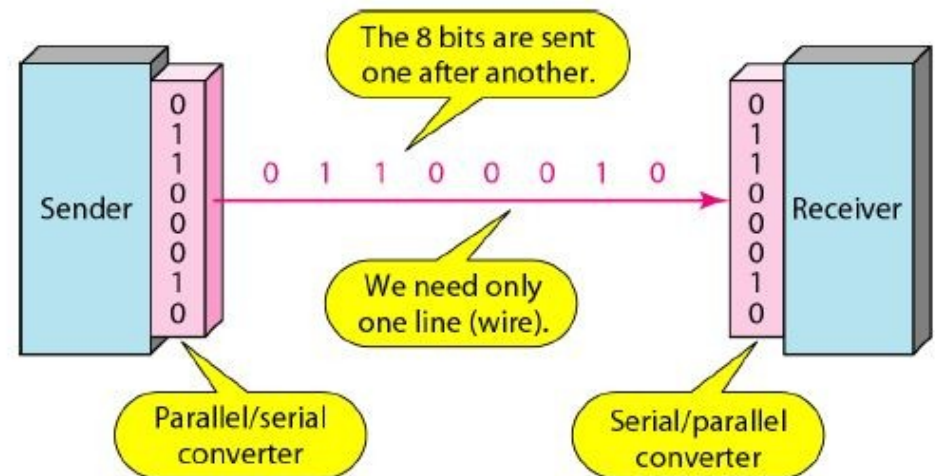
Parallel transmission

- Transmit groups of n bits each.
- Computers produce and consume data in groups of bits
- Use **n wires** to send n bits at one time
- Each bit has its own wire
- All n bits of one group can be transmitted with each clock tick from one device to another
- Advantage – Speed
- Disadvantage – Cost
- Limited to short distance



Serial Transmission

- One bit follows another
- Needs only one communication channel (wire)
- Cheaper
- Serial transmission occurs in one of three ways: asynchronous, synchronous, and isochronous

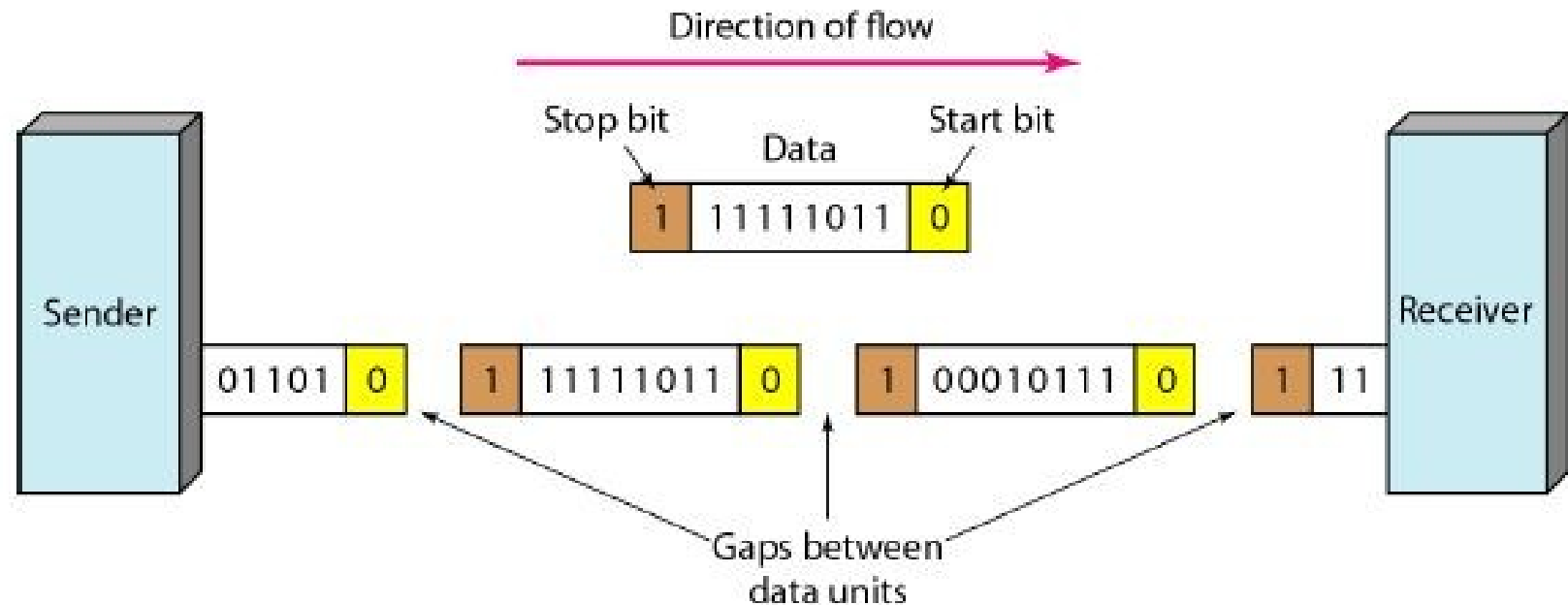




Asynchronous transmission

- Timing of a signal is unimportant
- Information is received and translated by agreed upon patterns (grouping the bit stream into bytes)
- The arrival of a new group is indicated with an extra bit added to the beginning of each byte
- When the byte is finished, 1 or more additional bits are appended to the end of the byte (stop bits)
- The transmission of each byte may then be followed by a gap of varying duration
 - Can be represented either by an idle channel or by a stream of additional stop bitss

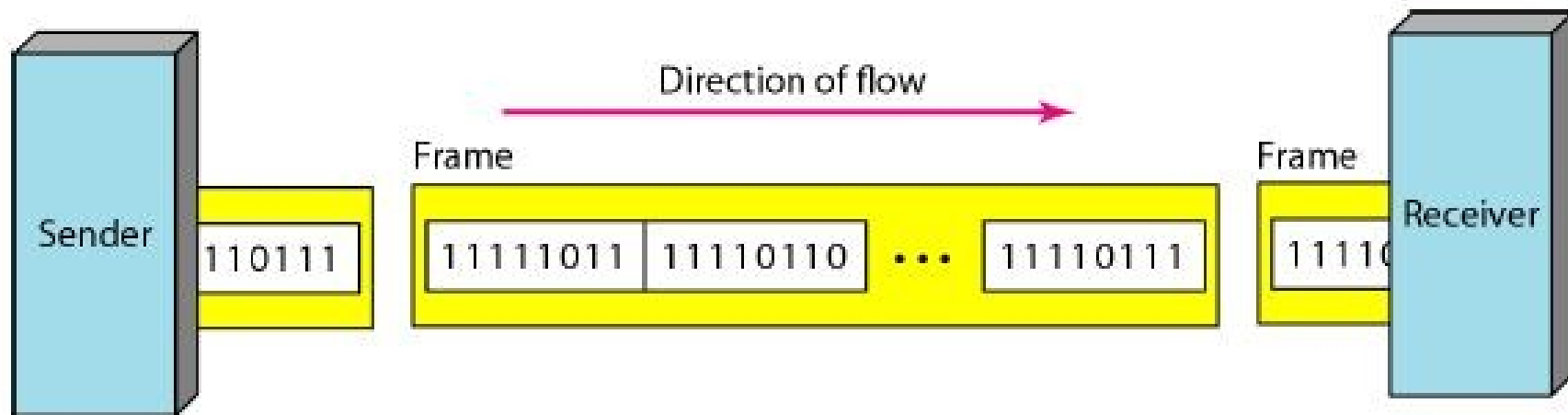
Asynchronous transmission



- Asynchronous here means "asynchronous at the byte level, but the bits are still synchronized; their durations are the same"

Synchronous Transmission

- The bit stream is combined into longer "frames," which may contain multiple bytes
- Each byte, however, is introduced onto the transmission link without a gap between it and the next one
- Data are transmitted as an unbroken string of 1s and 0s, and the receiver separates that string into the bytes, or characters, it needs to reconstruct the information



Next : Analog Transmission